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// Bluetooth HC-06/HC-05    <----> Arduino SENSor Shield V5.0 (4pin
TX,RX,-,+)
//      RX                  -NOI-          TX
//      TX                  -NOI-          RX
//      GND                 -NOI-          -
//      +5V                 -NOI-          +

// #include <LiquidCrystal_I2C.h> //Includes I2C-LCD1602 lib
#include <Wire.h> //Includes I2C Lib

#include "UCNEC.h"
int32_t temp = 0;
UCNEC myIR(A0);
#define F 16718055
#define B 16732845
#define L 16713975
#define R 16734885
#define S 16719075

// Motor control pins: L298N H bridge
const int enAPin = 6; // Left motor PWM speed control
const int in1Pin = 7; // Left motor Direction 1
const int in2Pin = 5; // Left motor Direction 2
const int in3Pin = 4; // Right motor Direction 1
const int in4Pin = 2; // Right motor Direction 2
const int enBPin = 3; // Right motor PWM speed control

int command;           //bien luu trang thai Bluetooth
int speedCar    = 220; // toc do 50 - 255.
int speed_Coeff = 2;   // he so suy giam <=> giam may lan toc do

void setup() {

    pinMode(enAPin, OUTPUT);
    pinMode(in1Pin, OUTPUT);
    pinMode(in2Pin, OUTPUT);
    pinMode(in3Pin, OUTPUT);
    pinMode(in4Pin, OUTPUT);
    pinMode(enBPin, OUTPUT);

    Serial.begin(9600); //Init Serial port rate set to 9600

    myIR.begin();
}

void loop() {
    if(myIR.available())
    {
        temp = myIR.read();
        Serial.println(temp);
    }

    if (temp == F){goAhead();}
}

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else if (temp == B) {goBack();}
else if (temp == L) {goLeft();}
else if (temp == R) {goRight();}
else if (temp == S) {stopRobot();temp=0;}

if (Serial.available() > 0) {
  command = Serial.read();
  stopRobot();          //Initialize with motors stopped.
  Serial.println(command);

  switch (command) {
    case 'F':goAhead();break;
    case 'B':goBack();break;
    case 'L':goLeft();break;
    case 'R':goRight();break;
    case 'I':goAheadRight();break;
    case 'G':goAheadLeft();break;
    case 'J':goBackRight();break;
    case 'H':goBackLeft();break;
    case '0':speedCar = 100;break;
    case '1':speedCar = 115;break;
    case '2':speedCar = 130;break;
    case '3':speedCar = 145;break;
    case '4':speedCar = 160;break;
    case '5':speedCar = 175;break;
    case '6':speedCar = 190;break;
    case '7':speedCar = 205;break;
    case '8':speedCar = 220;break;
    case '9':speedCar = 235;break;
    case 'q':speedCar = 255;break;

  }
}

}

void goAhead(){

  digitalWrite(in1Pin,LOW);
  digitalWrite(in2Pin,HIGH);          // LEFT tien
  digitalWrite(in3Pin,LOW);
  digitalWrite(in4Pin,HIGH);          // RIGHT tien
  analogWrite(enAPin, speedCar);
  analogWrite(enBPin, speedCar);

}

void goBack(){
  digitalWrite(in1Pin,HIGH);
  digitalWrite(in2Pin,LOW);           // LEFT lui
  digitalWrite(in3Pin,HIGH);
  digitalWrite(in4Pin,LOW);           // RIGHT lui
  analogWrite(enAPin, speedCar);
}

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        analogWrite(enBPin, speedCar);
    }

void goRight() {
    digitalWrite(in1Pin, LOW);
    digitalWrite(in2Pin, HIGH);           // LEFT tien
    digitalWrite(in3Pin, HIGH);
    digitalWrite(in4Pin, LOW);           // RIGHT lui
    analogWrite(enAPin, speedCar);
    analogWrite(enBPin, speedCar);

}

void goLeft() {
    digitalWrite(in1Pin, HIGH);
    digitalWrite(in2Pin, LOW);           // LEFT lui
    digitalWrite(in3Pin, LOW);
    digitalWrite(in4Pin, HIGH);         // RIGHT tien
    analogWrite(enAPin, speedCar);
    analogWrite(enBPin, speedCar);

}

void goAheadRight() { //vua tien - vua quay phai
    digitalWrite(in1Pin, LOW);
    digitalWrite(in2Pin, HIGH);         // LEFT tien
    digitalWrite(in3Pin, LOW);
    digitalWrite(in4Pin, HIGH);         // RIGHT tien
    analogWrite(enAPin, speedCar);
    analogWrite(enBPin, speedCar/speed_Coeff); //banh phai quay cham
hon
}

void goAheadLeft() {
    digitalWrite(in1Pin, LOW);
    digitalWrite(in2Pin, HIGH);         // LEFT tien
    digitalWrite(in3Pin, LOW);
    digitalWrite(in4Pin, HIGH);         // RIGHT tien
    analogWrite(enAPin, speedCar/speed_Coeff); //banh trai quay cham
hon
    analogWrite(enBPin, speedCar);

}

void goBackRight() { //vua lui - vua quay phai
    digitalWrite(in1Pin, HIGH);
    digitalWrite(in2Pin, LOW);         // LEFT lui
    digitalWrite(in3Pin, HIGH);
    digitalWrite(in4Pin, LOW);         // RIGHT lui
    analogWrite(enAPin, speedCar);
    analogWrite(enBPin, speedCar/speed_Coeff); //banh phai quay cham
hon

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    }

void goBackLeft() {
    digitalWrite(in1Pin, HIGH);
    digitalWrite(in2Pin, LOW);           // LEFT lui
    digitalWrite(in3Pin, HIGH);
    digitalWrite(in4Pin, LOW);           // RIGHT lui
    analogWrite(enAPin, speedCar/speed_Coeff); //banh trai quay cham hon
    analogWrite(enBPin, speedCar);

}

void stopRobot() {
    digitalWrite(in1Pin, LOW);
    digitalWrite(in2Pin, LOW);
    digitalWrite(in3Pin, LOW);
    digitalWrite(in4Pin, LOW);
    analogWrite(enAPin, 0);
    analogWrite(enBPin, 0);
}

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